MedTrack_DV

Hospital Operations & Patient Analytics Dashboard

Infosys Springboard Virtual Internship 7.0

Domain: Data Visualization

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Project Overview

MedTrack_DV is a healthcare analytics dashboard solution developed to analyze hospital operations, patient flow, department performance, and resource utilization.

The project integrates multiple healthcare datasets, perform data cleaning and transformation using Python, engineers operational KPIs, and presents the results through four interactive Tableau dashboards.

Final Output

A unified Tableau workbook(.twbx) containing four interconnected dashboards.

PROBLEM STATEMENT & OBJECTIVES

Problem Statement
Hospital datasets contain large amounts of patient and operational information. Without proper transformation and visualization, it becomes difficult to monitor hospital performance, understand patient movement, compare departments, and optimize available resources.

Objectives
Integrate multiple hospital and patient datasets.
Clean and transform raw healthcare data.
Develop reliable healthcare KPIs.
Analyze admissions, discharge and patient flow.
Compare department performance.
Monitor bed utilization and resource availability.
Build interactive Tableau dashboards.
Validate Tableau results against Python calculations.

Data Collection and Integration

Multiple datasets were collected and integrated to create a unified hospital analytics datasets.

Datasets Integrated

1. Patient
2. Admission
3. Department
4. Doctor
5. Employee
6. Bed
7. Disease
8. Patient Diagnostic
9. Insurance Provider
10. Patient Insurance

Dataset Scale

Datasets	Records
Patient	30,000
Admission	45,000
Doctor	98
Employee	500
Patient Diagnostic	63,269
Patient Insurance	21,617
Insurance Provider	50

85,724 records x 53 Fields

Data Cleaning & Transformation

The collected datasets were processed using python before being used in Tableau.

Cleaning Activities

- Removed duplicate records
- Handled missing values
- Standardized department information
- Integrated patient and admission records
- Connected department and resource information
- Created derived operational fields
- Prepared Tableau-ready data

Python Scripts

data_collection.py
Loads and verifies source datasets.

hospital_cleaning.py
Cleans and integrates healthcare data.

generate_hospital_kpis.py
Generates hospital and department KPIs.

KPI Engineering

Six major KPIs were developed as required by the project Specification.

KPI	Purpose
Total Admissions	Measures total hospital admissions
Average Length of Stay	Measures average duration of patient stay
Occupancy Rate	Measures occupied bed capacity
Readmission Rate	Identifies patients with multiple admissions
Bed Utilization Rate	Measures utilization of available beds
Department Efficiency Score	Compare department performance

Validation Approach
Python was used for KPI generation and Tableau calculated fields were used for dashboard implementation.

Dashboard Architecture

MedTrack_DV Dashboard Architecture

The project follows an end-to-end analytics workflow:

Data Collection → Data Cleaning & Transformation → KPI Engineering → Dashboard Planning → Tableau Dashboard Development → Dashboard Integration → Testing & Validation → Final Delivery

Four Dashboard Modules

1. Hospital Overview
2. Patient Flow
3. Department Analytics
4. Resource Utilization

Dashboard 1 : Hospital Overview

The Hospital Overview dashboard provides a high-level view of hospital performance.

KPIs

- Total Admissions
- Occupancy Rate
- Average Length to Stay
- Readmission Rate
- Bed Utilization

Visualizations

- Monthly operation/admission trends
- Department Efficiency Score
- Top 10 Insurance Providers by Patient count
- Hospital performance indicators

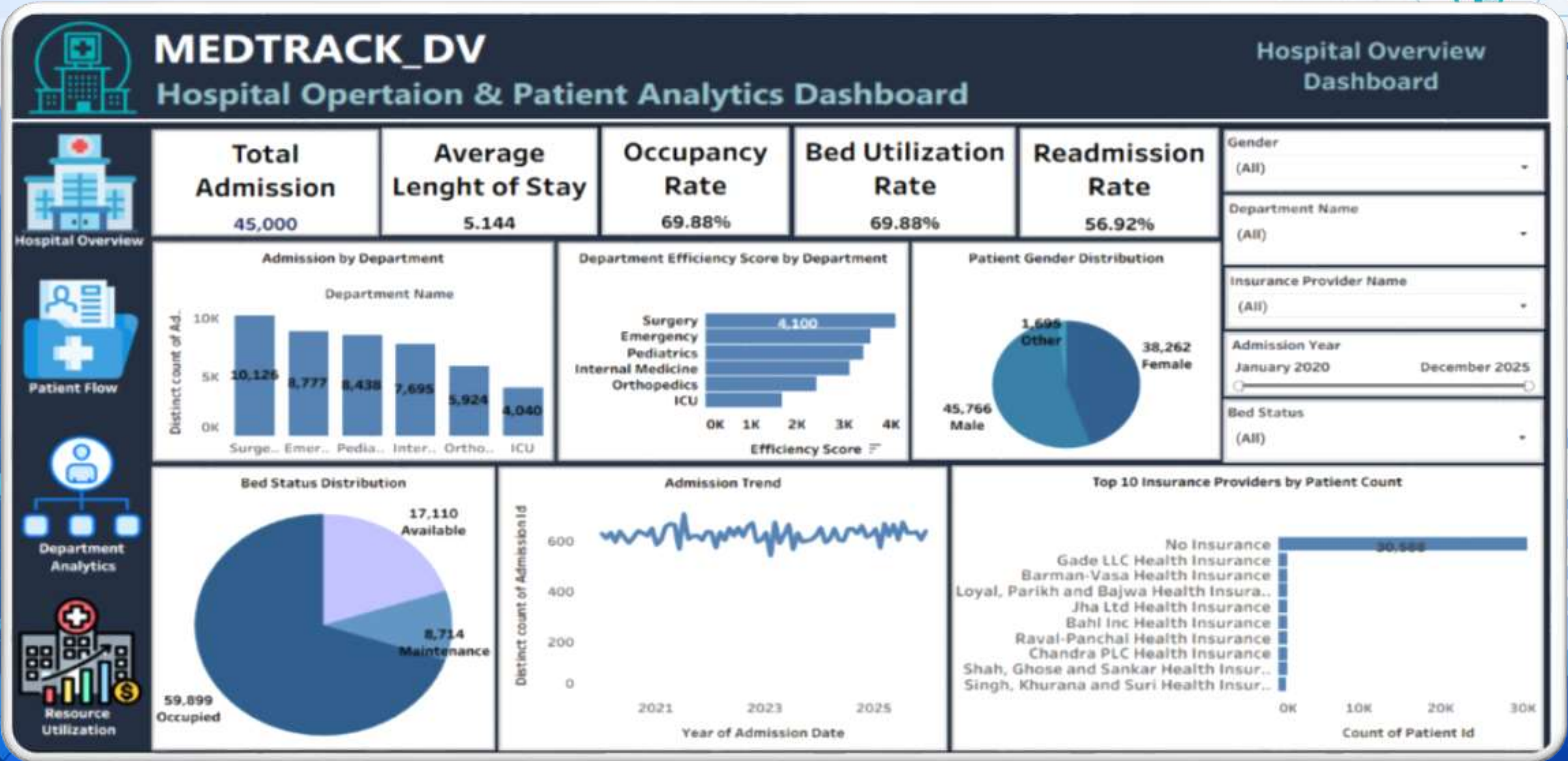
Interactivity

- Department filter
- Date /month filtering
- Interactive Charts
- Dashboard Navigation

Purpose

Provides Hospital administrators with a quick overview of operational performance.

Dashboard 1 : Hospital Overview



Dashboard 2 : Patient Flow

Patient Flow Analysis

The dashboard focuses on patient movement through the hospital.

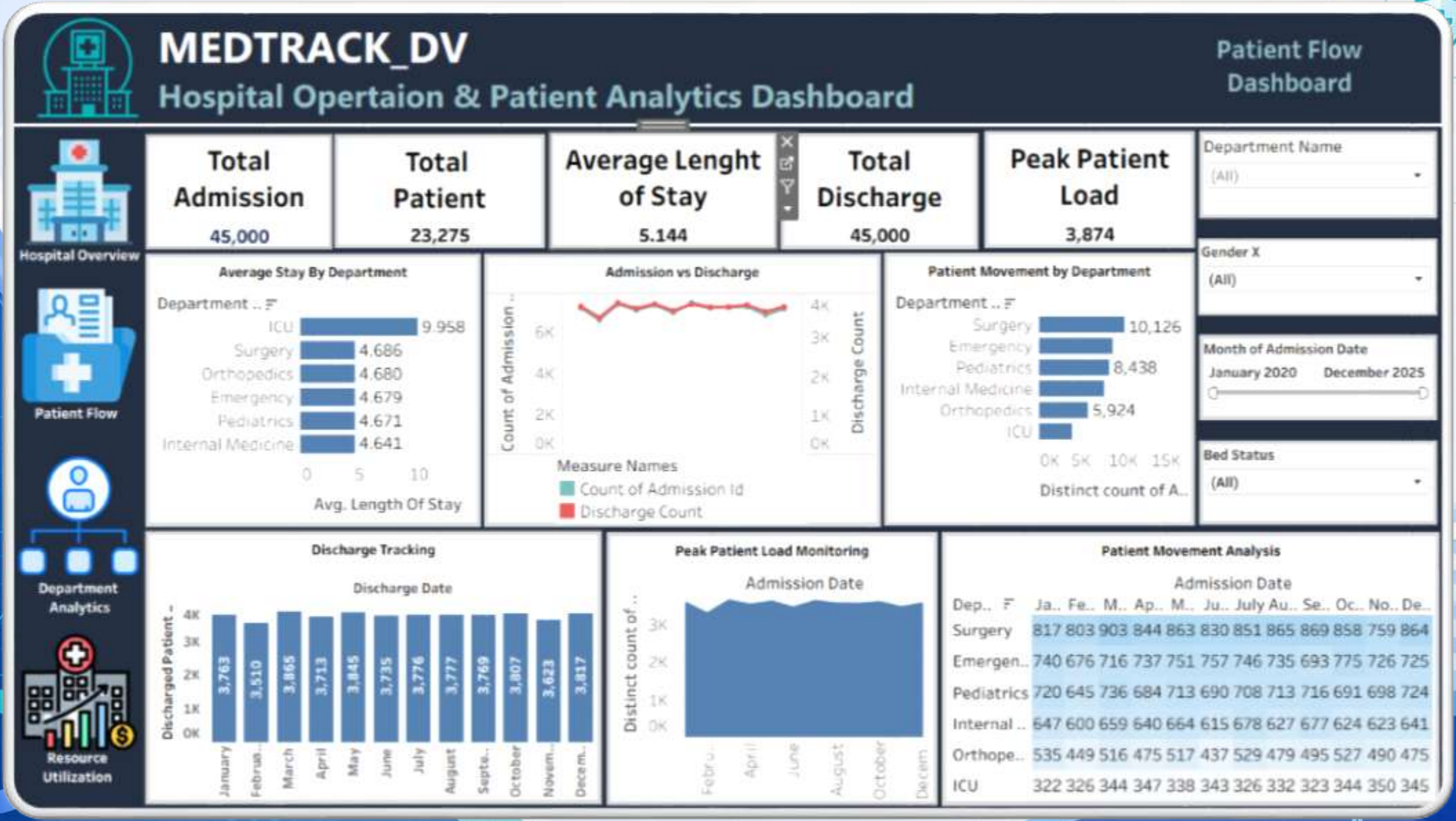
Key Analyses

Admission Trends	Discharge Tracking	Patient Movement Analysis	Average Stay analysis	Peak Patient Load Monitoring
changes in patient admission over time.	Compare admission and discharge activity.	Analyzes patient volume across departments.	Monitor Average patient Length of Stay.	Identifies periods with high patient activity.

Purpose

Helps identify Patient-flow patterns, high demand periods and potential operational bottlenecks

Dashboard 2 : Patient flow



Dashboard 3 : Department Analytics

Department Analytics
This dashboard provides department-level performance analysis.

KPIs

- Best Performing Department
- Highest Admissions Department
- Lowest Admissions Department

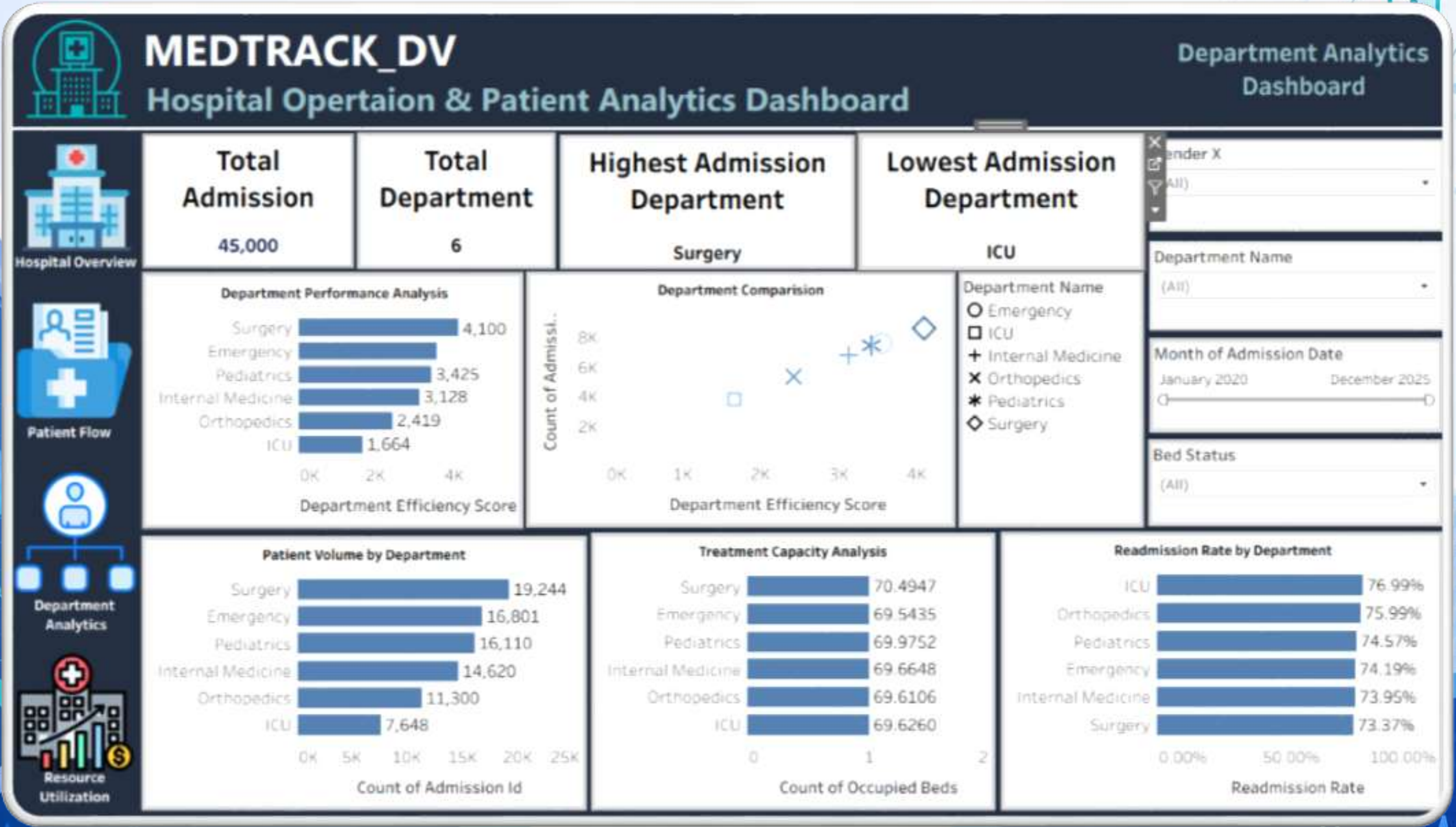
Visualizations

- Patient Volume by Department
- Readmission Rate by Department
- Department Efficiency Ranking
- Treatment Capacity
- Department Comparsion Scatter Plot

Purpose

- Analyzes department performance and identifies areas for improvement.

Dashboard 3 : Department Analytics



Dashboard 4 : Resource Utilization

The dashboard focuses on hospital capacity and resource usage.

KPIs

- Bed Utilization Rate
- Occupancy Rate
- Available Beds
- Occupied Beds

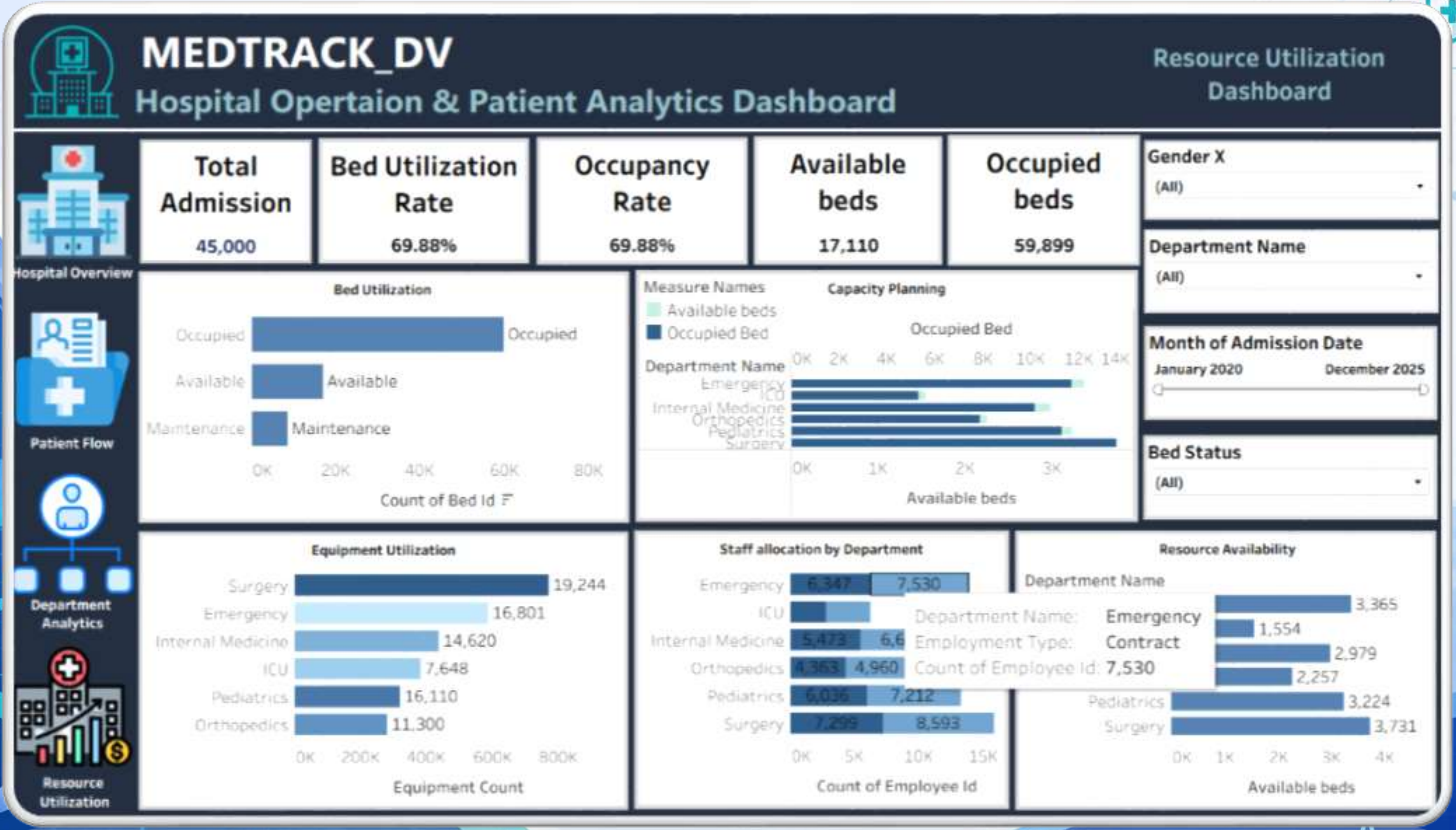
Visualizations

- Bed Utilization Analysis
- Staff Allocation Monitoring
- Equipment/Resource Utilization
- Capacity Planning
- Resource Availability Analysis

Purpose

Helps hospital management monitor bed capacity, resource distribution and utilization patterns for better operational planning.

Dashboard 4 : Resource Utilization



Dashboard Integration & Interactivity

The Four dashboard were integrated into a unified Tableau workbook.

Dashboard Navigation	➢ **Hospital Overview** ➢ **Patient Flow** ➢ **Department Analytics** ➢ **Resource Utilization**
Interactive Filters	➢ Department ➢ Date/Month ➢ Gender ➢ Bed status
Dashboard Action	➢ Interactive tooltips ➢ Department-level exploration ➢ Reset/Default dashboard view

Testing, Validation & Deliverables

QA Testing Performed

- Validated KPI calculations against python results.
- Checked admission counting and duplicate records.
- Validated readmission calculations at patient level.
- Tested bed utilization and occupancy calculations.
- Tested department-level calculations.
- Tested filters and dashboard interactions.
- Tested navigation between dashboards.
- Checked chart rendering and tooltips.
- Verified final Tableau workbook.

Testing Deliverables

- ❖ QA Checklist
- ❖ Dashboard Testing Report

Conclusion and Future Scope

Conclusion

MedTrack_DV successfully demonstrates and end-to-end healthcare analytics workflow:-

Raw Healthcare Data

- ✓ Data Cleaning & integration
- ✓ KPI Engineering
- ✓ Interactive Tableau Dashboards
- ✓ Testing & Validation

The Final solution provides insights into:

- ✓ Hospital performance
- ✓ Patient flow
- ✓ Department efficiency
- ✓ Bed and resource utilization

Future Scope

- Patient Admission forecasting
- Predictive readmission Analysis
- Bed-demand forecasting
- Real-time hospital data integration
- Staff workload forecasting
- Advanced resource optimization

Thank You

MedTrack_DV- Hospital Operations and Patient Analytics Dashboard